

Clinical Risk Assessment Report: Patient 92

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Positive for Metastasis
- Prediction Confidence: High Confidence
- Absolute Predicted Risk: 41.7% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 1.88x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as High-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Large Zone Emphasis (CT) (GLZLM_LGZE.CT), which elevated the patient's absolute risk by +5.8%. Biologically, this feature reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

2. Key Pathological & Imaging Drivers (Elevating Risk)

Large Zone Emphasis (CT) (GLZLM_LGZE.CT)	Value: -0.34 [Bottom 14%] (+5.8%)
<i>Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.</i>	
Long Run High Gray-Level Emphasis (PET) (GLRLM_LRHGE.PET)	Value: 2.23 [Top 4%] (+4.4%)
<i>Points to continuous, elongated bands of high-density or highly active tumor tissue.</i>	
75th Percentile Tumor Density (CONVENTIONAL_HUQ3)	Value: 0.69 [Avg (68th %ile)] (+3.3%)
<i>Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.</i>	
Tumor Pixel Correlation (CT) (GLCM_Correlation.CT)	Value: 1.59 [Top 3%] (+3.1%)
<i>Measures the linear dependency of gray levels; atypical values corroborate a complex, non-uniform spatial cellular arrangement.</i>	
Short Run Low Gray-Level Emphasis (CT) (GLRLM_SRLGE.CT)	Value: -0.25 [Bottom 4%] (+2.2%)
<i>Indicates fragmented areas of low density, often associated with diffuse micro-necrosis or loose tissue structure.</i>	
Small Zone Emphasis (PET) (GLZLM_SZE.PET)	Value: 1.07 [Top 13%] (+1.3%)
<i>Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.</i>	
Large Zone Low Gray-Level Emphasis (CT) (GLZLM_LZLGE.CT)	Value: 2.67 [Top 2%] (+1.2%)
<i>Indicates the dominance of large regions with low attenuation or metabolism, pointing to extensive necrosis or ground-glass opacities.</i>	
Minimum Metabolic Activity (SUVmin) (CONVENTIONAL_SUVbwmin)	Value: 2.1 [Top 5%] (+0.9%)
<i>Indicates the areas of lowest metabolic activity within the tumor, which may represent necrotic or poorly perfused regions.</i>	

3. Protective / Mitigating Factors (Reducing Risk)

No major risk-reducing features observed in the top drivers.

Machine Learning Feature Attribution (SHAP Decision Path)

